Hybrid Renewable Power Supply for Typical Public Facilities in Six Various Climate Zones in Morocco

Saad Ladide*‡, Amine EL Fathi**, Mohamed Bendaoud*, Hicham Hihi***, Khalid Faitah*

*LGECOS Laboratory, National School of Applied Sciences Cadi Ayyad University Marrakech, Morocco

** Faculty of Sciences and Techniques of Al Hoceima, Abdelmalek Essaadi University-Morocco

***National School of Applied Sciences, Sidi Mohamed Ben Abdellah University Fès, Morocco

(saad.ladide@gmail.com, amine.elfathi@gmail.com, Mohamed.bendaoud07@gmail.com, h.hihi@uca.ac.ma, k.faitah@uca.ac.ma)

‡ Corresponding Author; First author, BP 575 Boulevard Abdelkrim Al Khattabi, Guéliz 40000, Tel: (+212) 0524434745, Fax: (+212) 0524434740, saad.ladide@gmail.com

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Abstract- In this paper, a techno-economic analysis of a hybrid renewable energy system (HRES) is presented. This latter is covering the energy supply of typical facilities including public lighting, governmental institutions and school campus. In order to evaluate the impact of the geography towards the sizing of medium and small scale Microgrid, this study gives emphasis to six various climatic zones in Morocco. Based on HOMER Pro software, the study presents a detailed analysis with the objective of defining the suitable and sustainable solution. The assessment covers the economic feasibility, technical aspects such as the electricity generation and the excess of electricity. It also covers environmental indicators such as renewable energy fraction and greenhouse emissions among all systems for a project life of 25 years. Furthermore, a sensitivity analysis was carried out to evaluate the impact of various parameters on the viability of the proposed optimal systems. These parameters include the interest rate, the purchased power price and wind speed. In fact, the results show that for an interest rate of 8% and a power price of 0.13 $/kW, it is recommended that a grid-connected wind energy system is the appropriate solution for the cities located in the west of Morocco, while the grid-connected photovoltaic (PV) system is recommended for the cities located in the east except in Zagora where it is suitable to adopt a HRES with PV and wind energy. Likewise, the obtained cost of electricity in the different cities is comparable to the national electricity price 0.1 $/kWh.

Keywords HOMER Pro software, wind, solar, hybrid energy system, on-grid, optimization, sensitivity.

1. Introduction

Various studies have demonstrated the efficiency, inexhaustibility and high environmental protection of renewable energy resources [1][2]. Moreover, their costs are predictable and independent of fuel prices fluctuation [3][4][5][6]. However, those resources depend on many factors such as weather and climatic conditions which make them intermittent [7]. This character affects their reliability due the fact that one resource is not able to satisfy the load requirements all over the day [8]. All of these constraints show the need of the complementarity by using at least two energy resources [9][10]. The hybrid renewable energy system (HRES) combining PV and wind energy is considered as the most suitable alternative [11]. The recent advances in power electronic converters and renewable energy technologies have made HRES more popular in both grid-connected and off-grid mode [12]. Compared to conventional fossil fuels systems, HRES have the advantage of being more stable, environmentally friendly and present a low maintenance costs along their lifetimes. Several combinations of HRES can be adopted depending on the availability of the renewable energy resources. These combinations mainly had some obstacles in ensuring high reliability and efficiency of the HRES, which requires an optimization in the intent to get an optimal hybrid system able to satisfy the load requirements with a minimum capital investment [13][14]. Several research works reported that the economic feasibility depends on the location, while various studies have reported that the standalone HRESs are cost effective compared to grid-connected systems in remote areas [15].

In this perspective, HOMER Pro (Hybrid Optimization Model for Electric Renewable) software is considered as the most professional analytical tool for HRES, developed by the National Renewable Energy Laboratory (NREL), it can be used to optimally size the HRES. It ensures three principal calculations: simulation, optimization, and sensitivity analysis [16][17][18][19]. Indeed, it provides the user with technical and economic feasibilities of HRES. Also, it eliminates the configurations that do not meet the requirements. Moreover, all the optimized combinations are classified according to the total net present cost (NPC). According to the extracted data on hourly basis, the simulation phase is used to evaluate the cost effectiveness and the performance of the optimized system. HOMER Pro gives NPC and cost of electricity (COE) values of the optimized systems. In addition, a sensitivity analysis is considered as an important phase that gives opportunity to derive more concrete results by evaluating the impact of some critical variables [20]. In fact, Several studies have reported the optimization and sensitivity analysis of hybrid systems using HOMER Pro, e.g Rehman and Al-Hadhrami [21] studied a PV/diesel hybrid power system installed in a Saudi Arabian village to satisfy the electricity demand. The study showed that the system based only on four diesel generating units of 1500, 1000, 1750 and 250 kW with a diesel price of 0.2 $/l was found to be the most economical power system with a levelized cost of energy (LCOE) of 0.19$/kWh. This system became less economical when the fuel price exceeds 0.60$/l. The system including 21% of solar PV (2000 kWp) penetration, four diesel generators of 1250, 750, 2250 and 250 kW a battery bank and a power converter of 3000 kW with a COE of 0.219$/kWh was considered as the second best economical configuration.

Mahmoud and Ibrik [22] noted that the PV/diesel hybrid system is cost effective than the basic diesel system for remote areas in Palestine. Adaramola investigated the integration possibility of hybrid power systems for electricity generation in the north of Nigeria [23]. The study showed that PV/diesel/Battery hybrid system presents the economical combination for standalone systems.

Aalm and Sadrul assessed PV-diesel and wind-diesel energy systems for Kutubdia islands in Bangladesh by using HOMER software and RETScreen simulation tools [24]. The HOMER software was used to optimize the system configuration while RETScreen was used for the analysis and the evaluation of the COE and the CO_2 emission. The study results in an equity payback of 5.4 years with a CO_2 emission reduction of 54.3 t/year for the PV-diesel system. While in the wind-diesel system, the equity payback was assumed to be 11.2 years with an annual CO_2 emission reduction of 42.9 t/year. Furthermore, the PV-diesel system was found to be the cost effective with a COE of $0.353/kWh compared to wind-diesel system (0.487 $/kWh).

Olatomiwa et al examined the economic feasibility of HRES including PV, wind, and diesel resources in Nigeria [25]. The results show that the PV/wind/diesel/battery system reports the optimal configuration for rural health clinics (RHC) applications in Jos, Sokoto, Maiduguri and Enugu with a COE($/kW) of 0.324, 0.336, 0.391 and 0.449 respectively. While the PV/diesel/battery configuration is considered as an optimal configuration for RHC in the remote areas of Iseyin and Port-Harcourt with a COE($/kW) of 0.476 and 0.482 respectively. This can be justified by the high potential of these locations in terms of solar energy. The study also shows the high cost of the hybrid system including diesel generator due to its high COE ($0.911/kWh) and its CO_2 emissions that exceed 9.2 t/year for all the considered sites. In another research work, Olatomiwa et al investigated technically and economically the introduction of solar and wind energy to supply the power demand of mobile base transceiver station (BTS) in Nigeria [26]. The study was based on the analysis of the COE, the NPC and the CO_2 emissions. The results report that the PV- diesel-battery system is considered as the best configuration compared to the conventional system based only on the diesel generator. Comparatively to the standalone diesel generator system, the proposed HRES presents a low COE of 0.4 $/kW with an emission reduction of 16.4 t/year. The same authors analyzed the feasibility of HRES, including PV, wind turbine and diesel generator for six rural locations in Nigeria [26]. The study was based on the analysis of the NPC, COE, renewable fraction (RF) and the CO_2 emissions. The results showed that the PV/diesel/battery system was cost effective compared to the conventional diesel generator that has a high COE ($1.075/kWh), with 58.4 t/year of CO_2 emissions.

M. Kamran et al presented a design and optimization of Standalone HRES for rural areas located in Punjab, Pakistan [27]. The study proposes three strategies, the first one uses the diesel generator integrated with solar and wind power systems, this latter is considered as the most expensive with a total NPC of $ 670,121 and a COE of $ 0.0936/kWh. The recovery period is assumed to be 13 years. The second strategy presented a renewable fraction of 100% with a minimized (NPC) of $ 479,835, the COE reduces to $ 0.0670/kWh and a recovery time of 5.5 years. The last one enacts renewable energy systems with capacity shortage scheme. NPC and the COE reduces to $ 284,877 and $ 0.0437/kWh respectively compared the previous strategy.

A study in Nigeria was carried out to design an off-grid HRES based on PV and battery to supply the load demand of registration centers located in Nkanu-West Kaura, Nigeria [28]. The results demonstrated that the most optimized PV-Battery power system for an energy consumption of 2.3kWh/day with a 235W peak demand load consists of 0.9 kW solar PV array, 9 Hoppecke 24 OPzS 300 Battery and a 1kW DC/AC inverter.

Srivastava and Giri analyzed a HRES designed to cover the load demand of Measurement & Instrument Laboratory in Electrical Engineering Department of Madan Mohan Malaviya University of Technology at Gorakhpur, India [29]. The results showed that the system based on PV, DG with battery and inverter is considered as the most economical system with an initial cost of 9334$, an operating cost of 320$/year and a COE of 0.262$

Maatallah et al [30] investigated the techno-economic feasibility of hybrid power system consisting of PV, wind and DG in the northernmost city in Africa, Bizerte, Tunisia. The study reports that the proposed hybrid system is cost effective with a low COE of 0.26 $/kWh. Moreover, the analysis shows that a configuration with batteries will reduce the CO_2 emission and the NPC by 81%, 85% and 29% respectively.

A study conducted in Saudi Arabia was carried out to propose an optimal hybrid system based on PV/wind turbines, converter, batteries, electrolyser, fuel cell (FC) and hydrogen tank for different area [31]. The study showed that the combination including three wind turbines, seven batteries storage bank, a PV field of 2 kWc and a converter of a rated power of 2 kW presents the most suitable alternative due to its low COE of 0.609 $/kWh.

Ye et al presented a feasibility analysis of a HRES based on the renewable energy availability and the local electricity demand estimation in 2020 by using HOMER Pro software [32]. The results show that the COE for the grid, grid-connected hybrid RE, standalone hybrid RE is 0.127, 0.033 and 0.123 $/kWh, respectively. Furthermore, the study resulted that the proposed grid-connected hybrid system based on natural gas and wind power is the most economic and environment friendly. Indeed, the carbon emissions will be reduced by a rate of 40 to 70% compared to the electricity generated from the existing power plants.

Sharafi et al investigated the optimization approach for the hybrid renewable energy system sizing for a residential building including heating, cooling, hot water, electrical and transportation loads [33]. The obtained results have shown that the renewable energy ratio is about 100% for a configuration including 200 kW biomass boiler, 73 kW wind turbine and a plug-in electric vehicle. The optimized configuration gives an NPC of $705,180 and 2.4 t/year as a total CO_2 emission.

Sigarchian et al performed a techno-economic feasibility of a hybrid energy system combining PV, biogas and wind turbine to feed electricity to a Kenyan village [34]. The average and the peak load of the village are around 8 kW and 16.5 kW respectively. The results report that the hybrid system based on biogas is suitable because it saves 17 tons of CO_2 per year and presents a COE of 0.25 $/kWh compared to the system based on diesel generator that has a COE of 0.31 $/kWh.

Salehin et al presented both, optimization and economic viability of an existing grid-connected system based only on DG that feed electricity to a remote village in Algeria [35]. The study proposes the add of wind turbines to the current power system to minimize the diesel consumption and reduce the CO2 emissions. The results show that for a wind speed upper than 5.48 m/s and a fuel price of 0.162 $/l, the based wind–diesel hybrid system becomes feasible. In Ethiopia, Bekele and Tadesse investigated the feasibility of a hybrid power system based on PV, wind and hydropower to meet the electricity demand of six different regions [36]. The proposed hybrid system presents a competitive COE less than $0.16/kWh for the region of TABA compared to the Ethiopian tariff.

Due to the energy deficit in Morocco, and given the country's willingness to invest in renewable energies, the Moroccan government established several legislatives laws aiming to promote renewable energy production. In addition, Morocco has managed to increase the share of renewable energy in the energy mix by installing several power plants based on wind, CPS and photovoltaic energies. In order to follow this vision, it is also necessary to conduct several techno-economic studies and investigations for small and medium scale projects. The optimal sizing of HRES improves significantly their economic and technical performances. This optimization can boost the extensive use of such eco-friendly systems. To perform the techno-economic feasibility of HRES configurations, optimization and sensitivity analysis need to be performed. So far, to the best knowledge of the authors, there is no comprehensive works on techno-economic assessment of PV/Wind hybrid system that has been performed in Morocco for typical public facilities. In this context, a techno-economic investigation of grid-connected HRES, designed to meet the electrical energy demand will be conducted. Considering that Morocco has different climate zones, six cities located in different areas in Morocco will be considered. The main contribution of this study is to assess, via a techno-economic analysis of a hybrid system connected to the grid, the deployment of small and medium Microgrid based on solar and/or wind resources. It is also an opportunity to show the advantage of making electricity production closer to the consumption, limiting investments in transmission infrastructure and reducing losses in distribution networks. Besides the economic feasibility and their associated indicators, this study covers also technical aspects such as the electricity generation and the excess of electricity. It also covers environmental indicators such as renewable energy fraction and greenhouse emissions among all systems for a project life of 25 years. Furthermore, a sensitivity analysis will be carried out to evaluate the impact of various parameters on the viability of the optimal systems. These parameters include the interest rate, the purchased power price as an economic parameter and the wind speed variation as a technical sensitive input.

This paper will be organized as follows. Besides the introduction, the next section describes the different components of the hybrid system, the load demand of typical facilities for the six cities and their associated model inputs. Then, the obtained optimization results will be presented and discussed. In the last section, a sensitivity analysis and its effect on the viability of the optimal proposed systems will be discussed. Fig. 1 shows the adopted methodology in Homer software. In fact, after the choice of the six locations, the load demand, the RE resource assessment (PV and wind), the equipment selection and the economic boundaries have to be specified accurately.

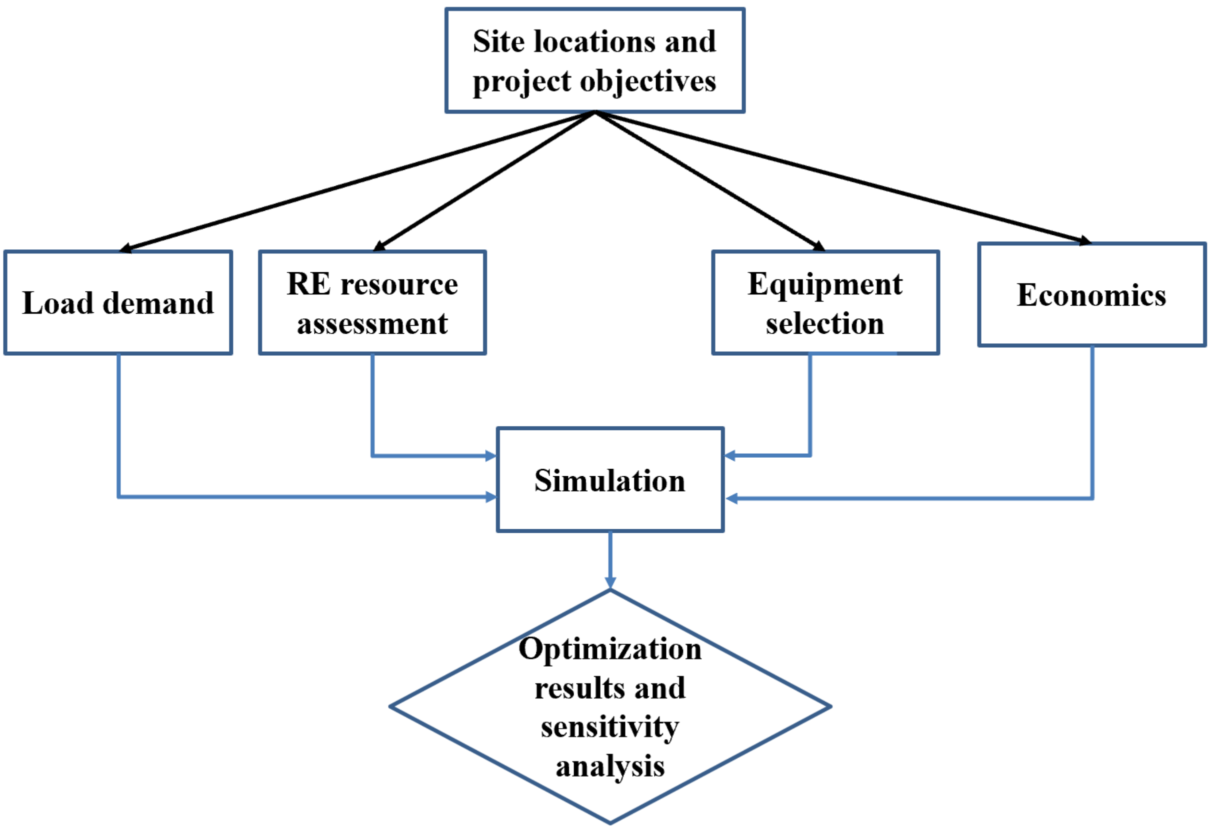


Fig. 1. Adopted methodology in Homer software

Nomenclature

HRES	Hybrid renewable energy system	RE	Renewable Rate
WT	Wind turbine	IR	Interest Rate
E_{pv}	Energy (kW h)	DG	Diesel Generator
A’	Total solar panel area (m²)	GHI	Global Horizontal Irradiance
P	Solar panel yield (%)	GHG	Green House Gas
C_p	Performance coefficient	RES	Renewable Energy System
P	Density of air	λ	Tip speed ratio
PR	Performance ratio	β	Blade pitch
CI	Clearness Index	HOMER	Hybrid Optimization of Multiple Electric Renewables
i	Real Discount Rate	A	Swiped Area(m2)
i’	Nominal Discount Rate	ηt	Wind turbine efficiency
CS	Capacity Shortage	ηg	Generator efficiency
TNPC	Total Net Present Cost	OC	Operating Cost
Dmap	Data Map	O&M	Operation and Management
		COE	Cost of Energy

2. Sites Specifications

2.1. Sites location

Morocco is an African country characterized by its climatic diversity which includes continental, oceanic, Mediterranean, semi-arid and Sub-Mediterranean climate. This study will consider 6 Moroccan cities covering the different climatic regions as shown in Fig. 2. Table 1 shows the studied cities each with its coordinate and climate zone.

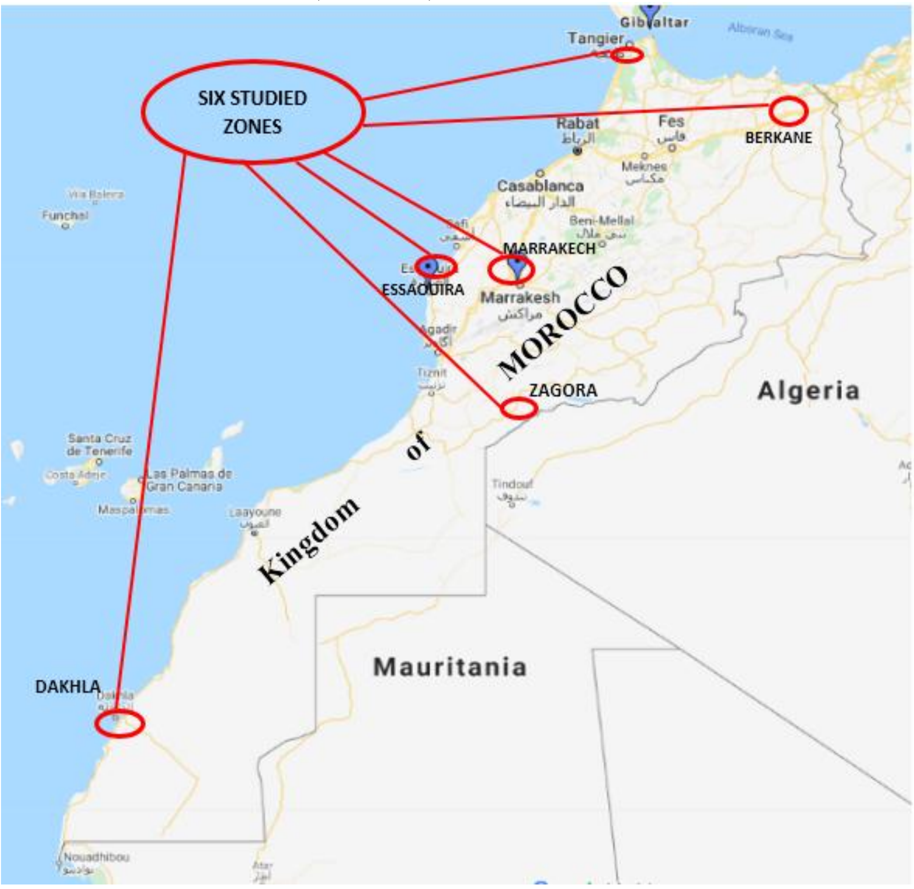


Fig. 2. Map of Morocco showing the studied locations

Table 1: Studied climatic zones

Climatic zone	City	Climate	Coordinates
Zone 1	Tangier	Mediterranean	35°46′ 0″ N, 5°48′ 0″ W
Zone 2	Marrakech	Semi-arid	31°37′ 48″ N, 8°0′ 32″ W
Zone 3	Berkane	Semi-arid	34°55′03.41″N, 2°19′02.89″W
Zone 4	Essaouira	Oceanic	31° 30' 45.00" N, 9° 46' 12.00" W
Zone 5	Dakhla	Arid	23° 43′ 0″ N, 15° 57′ 0″ W
Zone 6	Zagora	Warm desert	30°19′50″N 5°50′17″W

2.1. Resources potential

As shown in Fig. 3, Morocco benefits from favorable geographic and climatic conditions. It has a high potential of solar power generation with an abundant solar resource (a potential of 2600 kWh/m2/year), as well as a good wind energy potential with 17 regions that have been selected as suitable for wind power generation [45]. Morocco has 3500 km of coastline where the average wind speeds can reach up to 10 m/s. It is therefore estimated that the total theoretical Moroccan wind power potential is about 25 GW [37][38].

2.1.1. Wind

The wind speed data were obtained from the National Aeronautics and Space Administration (NASA). As illustrated in Fig. 4, the obtained data show that Marrakech has a minimum wind speed ranging from 2.2 to 3.3 m/s, while the maximum speed can be noticed in Tangier with an annual average wind speed of 7.7m/s. Based on the wind speed, the energy generated by the wind turbine can be expressed as [39], [40]:

$$P_{wt} = \frac{1}{2}\rho.A.V.Cp(\lambda,\beta).\eta_t.\eta_g \quad (1)$$

Where ρ=1.22 kg/m^3 is the density of air, Cp is the performance coefficient of the turbine, λ is the tip speed ratio, β is the Blade pitch angle as 0°, A is the swept area by the wind turbine (m^2), V is the wind speed (m/s), η_t is the wind turbine efficiency and η_g is the generator efficiency.

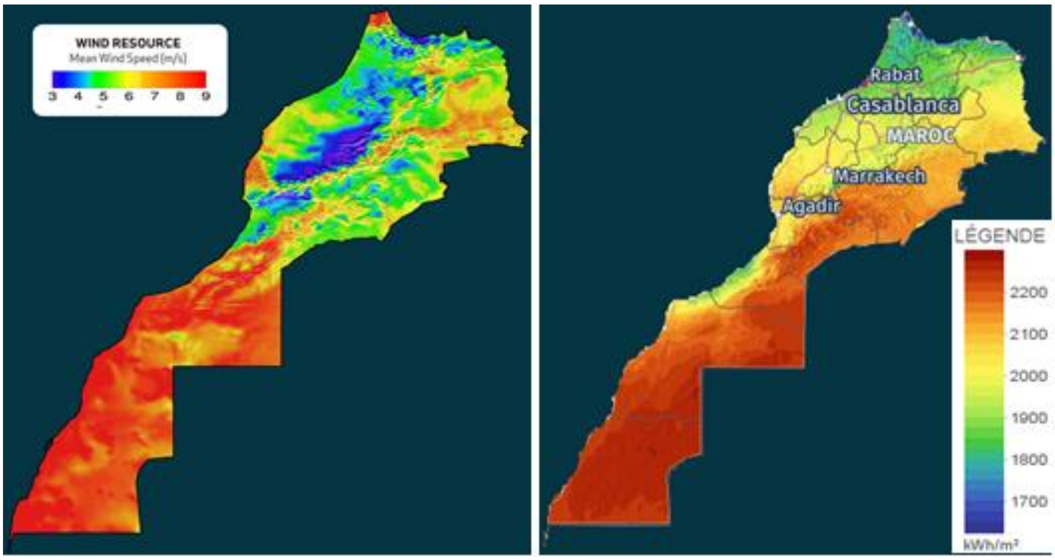


Fig. 3. Solar radiation (kWh/m2/day) and wind speed (m/s) area of Morocco with highest potential for solar and energy

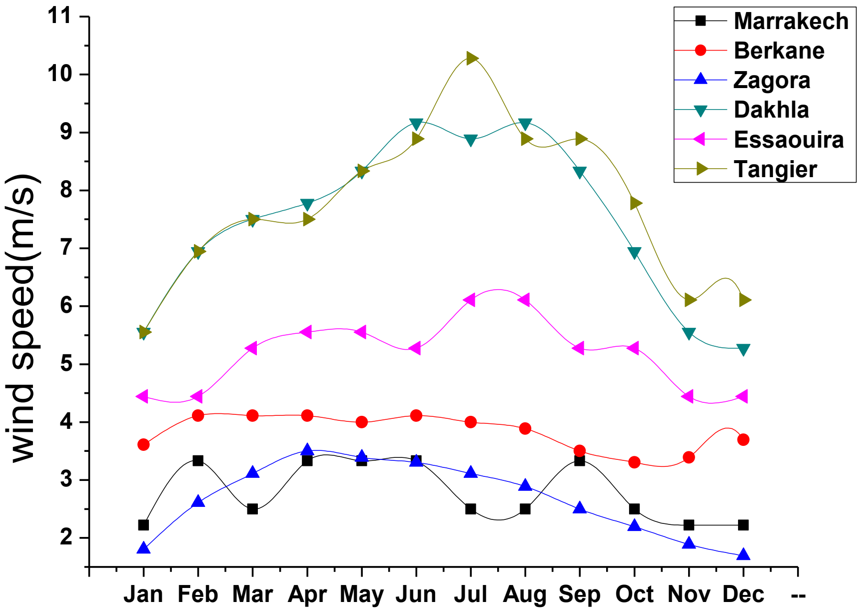


Fig. 4. Monthly Average wind speed

2.1.2. *Solar*

The global solar radiation data were obtained from NASA's website and illustrated in Fig. 5. From this figure, the solar radiation varies from one city to another, with a maximal average registered in the months of April and September. The solar energy output can be calculated based on the solar radiation as follow as:

$$E_{pv} = H * P * A' * PR \quad (2)$$

Where, E_{pv} is the solar energy (kWh), A' is the total solar panel area of the photovoltaic generator (m²), P is the solar panel yield (%), H is the annual average solar radiation and PR is the performance ratio.

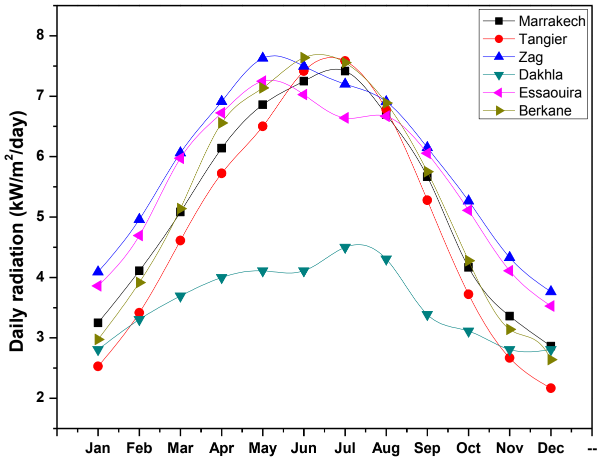


Fig. 5. Monthly Average radiation

3. Load Demand and System Components

3.1. Load demand

The electric load taken in this study is listed in *Table 2*. It reports a typical load demand which can be divided in three main categories:

- Public lighting that covers the lighting of 4 streets in the city center. They operate for a half day and the consumed energy is between 165.2 and 196 kW/day.
- Governmental institutions including school, administrations, police office as follow: School consisting of 15 classes, 4 offices and one library, consuming an amount of energy of 29.6kWh/day in winter and 1.6 kWh in the period between July and august due to the stopping of studies in this period. In addition, the administration is composed of 6 offices, consuming an amount of energy of 28.2kWh/day.
- The school campus that hosts 100 students consumes 23.44 kW/day (September-June) and 3.6 kWh/day in the period between July and August.

After the load determination and based on the normal distribution, the HOMER Pro software allows drawing the hourly perturbation factor with a mean of zero and a standard deviation equal to the considered input value "hourly noise". For This study, appropriate values of the daily and hourly perturbation factors are taken as 5% and 15%, respectively. The presented load profile will be adopted for the entire study in intent to assess its impact on the proposed HRES sizing for the selected cities.

Table 2: Summary of different equipment

Load specification			**Season**					
			September-June				**July-August**	
			Quantity	**Power/unit**	**Working time(h)**	**Energy amount(kWh /day)**	**Working time(h)**	**Energy amount(kWh /day)**
Governmental institutions	***school***	***lighting***	20	80	3	4.8	1	1.6
		PC	40	80	4	12.8	0	0
		library	10	100	5	5		
	administrations	***lighting***	12	80	5	4.8	5	4.8
		PC	16	80	5	6.4	5	6.4
		A/C	8	1100	2	17.6	4	35.2
	police office	***lighting***	10	80	5	4	4	3.2
		PC	4	80	5	1.6	4	1.6
		A/C	2	1100	2	4.4	4	8.8
school campus	***Lighting***		30	80	6	14.4	5	9 3.6
	Offices	***Lighting***	2	80	4	0.64	0	0

	A/C	2	1100	2	4.4		
	Library	10	100	4	4		
	Additional facilities		100	3	0.3		
Public lighting	50		200	11	110	10	100
Total					196		165.2

Fig. 6 illustrates the data map of the monthly average load demand. It can be noticed from this figure that the electric consumption has its maximum in the period between September and June with an average daily consumption of 219.05 kWh/day and 19 kWh as a daily peak. The yearly data map shows that the demand increased in the evening due to public lighting, which consumes more than the other loads.

3.2. HOMER pro and System components

As mentioned above, HOMER is considered as one of the main software used for HRES sizing and optimization; it performs three main tasks: simulation, optimization, and sensitivity analysis [41]. HOMER models the performance of a particular hybrid system configuration each hour of the year to define its technical feasibility and life-cycle cost. A feasible system is presented as a hybrid system configuration that can satisfy the required load and meeting other constraints imposed on the system. The optimization process is carried out to simulate several system configurations and find the system that satisfies the technical and economic constraints at the lowest life-cycle cost. In the sensitivity analysis process, several optimizations under a range of input values are performed with the intent to investigate the effects of uncertainty or changes in the model inputs (such as changes in wind speed or diesel price). [42][43]. Fig. 7 presents the schematic diagram of the hybrid system considered in this study with possible combinations of the following components: wind turbine, PV, converter, diesel generator, and battery system.

3.2.1 PV system

The solar panel cost depends on many factors including size of the solar panel, technology and the brand. The Trina Tall max1500V PV panel is considered in this study with a 72 mono-crystalline cells , while its power output is between 305 and 315 watts under Standard Test Conditions (STC) [44].

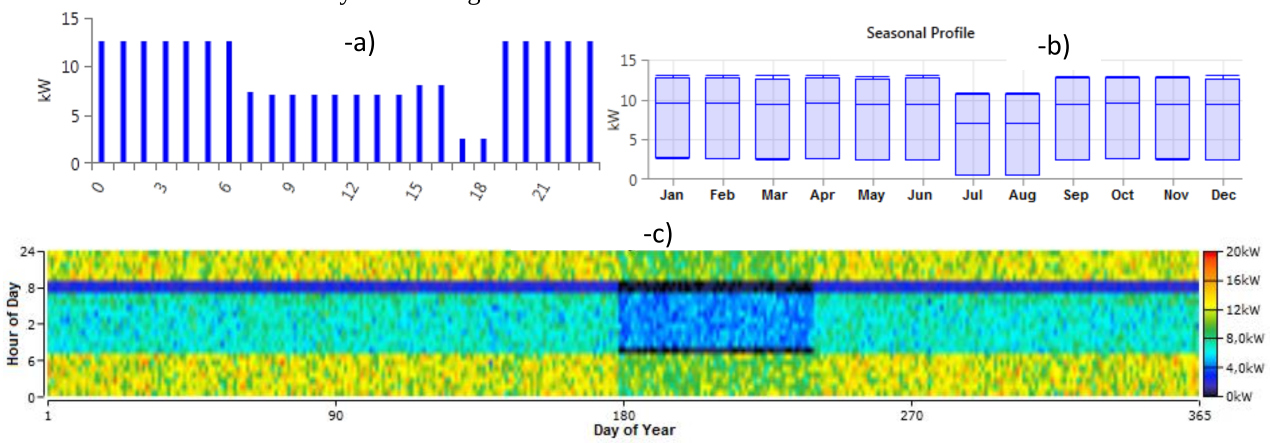


Fig. 6. (a) Daily, (b) seasonal and (c) yearly Load profile

Table 3: Technical details of PV and wind components

PV system		***EOC wind turbine***	
Rated power (W_p)	*315*	*Design standard*	*IEC 61400-2 Wind class III*
Maximum Power Voltage (V)	*37.1*	*Cut-In wind speed*	*2.5 m/s*
Maximum Power Current (A)	*8.51*	*Cut-out wind speed:*	*20.0 m/s*
		Rated Power	*20 kW*
Open Circuit Voltage (V)	*45.6*	*Rotor Diameter*	*15.8 m*

Short Circuit Current (A)	*9*	*Alternator*	*3-Phase, Permanent synchronous permanent*
Derating factor (%)	*80*	*Blades*	*3*
Temperature coefficient (%/°C)	*- 0.41*	*Tip speed*	*4.4 m/s*
Operation temperature (C)	*44*	*Life time*	*20 years*
Lifetime (Years)	*25*	*Swept area*	*196.3 m²*
Efficiency (%)	*16.2*	Gear box	*without*

The technical specifications are cited in Table 3. The cost of PV panel in Morocco varies between 1 and 1.5 $/Wp, whereas the annual maintenance cost is taken as $10[45]. The PV panel lifetime is defined to be 25 years. Moreover, the ground reflectance and the de-rating factor are considered as 20% and 80%, respectively. Also the temperature affects the output power production, whereas the temperature effect on power is assumed to be -0.41 %/°C and the nominal operating temperature is defined as 44°C [44].

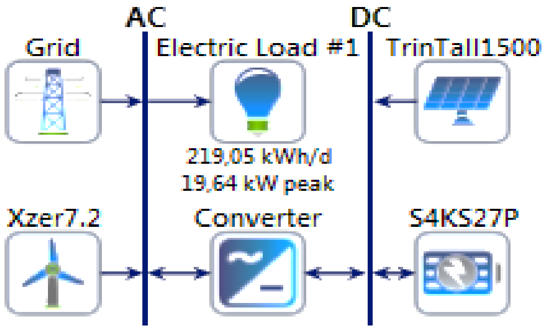


Fig. 7. HRES schematic diagram

3.2.2 Wind turbine

The wind turbine model selected for this study is EOCYCLE EO20, it is considered as one of the most efficient rotor designs tested by the National Renewable Energy Laboratory, and it has as rated power of 20kW. The technical specifications of the wind turbine are reported in Table 3 . The lifetime and the height of this equipment is assumed to be 20 years and 30 *m* ,respectively [46]. Furthermore, the cost of the wind turbine is $30000, whereas the replacement and the annual maintenance costs are taken $27000 and $200, respectively [47]. Fig. 8 shows the wind power curve of an EOCYCLE EO2 wind turbine.

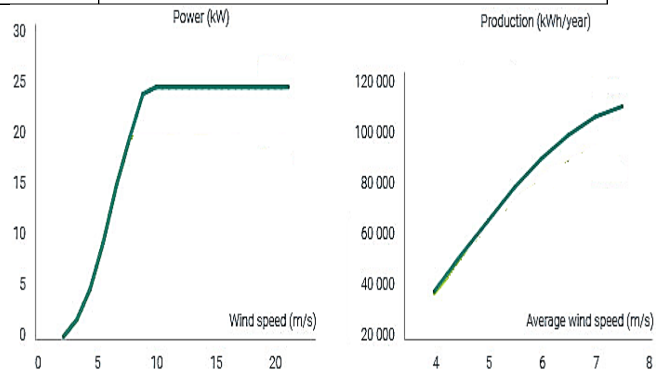


Fig. 8. Wind turbine power curve

3.2.3 Converter

Based on various research works, the cost of converters can be taken as $300/kW, whereas the replacement and the annual maintenance costs are taken as $27000 and $10, respectively [48]. The converter lifetime was assumed to be 15 years and its efficiency was assumed to be 90% [49].

3.2.4 Grid

In this study, a grid-connected system will be adopted. According to the National Office of Electricity (this office is responsible for the provision of electricity as well as the operation of the national grid), the purchase price of electricity is assumed to be $0.14/kW. On the other hand, the selling possibility started after the adoption of the law 58-15 by the government in the end of 2015. This law allows private entities which are connected to high and medium voltage network to sell the excess of the produced energy from renewable sources. In this study, the selling cost is fixed in its minimal value of $0.03/kW. On the other hand, if the connection is planned to be done on the medium-voltage network, other costs have to be considered such as, the cost of the transformer (220V to 22,000V), the pylons, the wires, the protection components and the labor, etc. According to the local market prices in Morocco, in order to extend the network for 1 km to reach the nearest point of the medium-voltage network, it is expected to spend $25,000.

3.2.5 Batteries

Despite the fact that a gird-connected system is adopted in this study, the effect of using batteries will also be evaluated during the simulation/optimization process. In this context, the Rolls Surrette 4 KS 27P battery was selected for this study; it is dedicated to large systems with a capacity of 1460 Ah. Its lifetime is assumed to be more than 10 year depending on the maintenance efficiency. The 4 KS 27P is a lead acid battery, the cost of the battery unit is $1500, while the replacement and the annual maintenance cost are $15,000 and $75, respectively [50].

4. Results and Discussion

In this paper, several scenarios have been investigated by using HOMER Pro which is carried out to provide all possible configurations including different technical and economic aspects. The proposed HRES is designed to meet the load demand throughout the 25 years. The system must meet the requirements given the changes that are practically affecting the HRES such as the evolution of the load demand and the different component costs. In fact, the HRES was evaluated by performing a sensitivity analysis to assess the impact of some variables such as the load demand as well as the evolution of the purchase price of electricity on the optimal design. As a result, the HOMER Pro ranked the optimal combinations which are assumed to continuously meet the load demand according to their NPC, RF and COE.

4.1. Electricity generation

The monthly average of the produced electricity by the different components in each city is presented in Fig. 9. The simulation results show that the generated power from wind turbines is the only source of energy in Dakhla and Tangier with low values of purchased electricity compared to the other cities; this is due to the fact that these two cities enjoy a high level of wind speed. Due to its high level of solar irradiation and low values of wind speed, the PV energy is the only source of electricity in Marrakech. In this case, the generated electricity decreases in night with the absence of solar irradiation which explains the considerable amount of purchased electricity. For the cities of Essaouira and Zagora, the optimal system combines PV and wind energy with purchased electricity from the grid. As illustrated for these two cities, the amount of purchased electricity is greater than the amount of the produced energy by the PV and the wind turbines. It can be noted that the purchased electricity has a high amount in Marrakech, Essaouira and Zagora, which is approximately between 40 and 50% compared to the other two cities that not exceed 20% as electricity penetration.

Table 4: Sensitivity analysis results

City	Sensitivity		Architecture				System cost			Energy production(kW/year)				
	Purchased power price ($)	Interst rate IR (%)	EOS	PV (kW)	Converter (kW)	Battery	COE ($)	Operating Cost ($)	Initial capital ($)	EOS	PV	Battery	Grid Purchased	Grid Sold
	0,08	8	0	42	33	0	0,1	2840	114510	0	8379	0	54466	54187
Marrakech	0,08	9	0	42	33	0	0,11	2849	114560	0	83769	0	54466	54187
	0,08	10	0	42	33	0	0,12	2839	114561	0	83769	0	54466	54187
	0,13	8	0	42	33	0	0,12	5583	114561	0	83769	0	54466	54187
	0,13	9	0	42	33	0	0,13	5573	114561	0	83769	0	54466	54187
	0,13	10	0	42	33	0	0,14	5562	114561	0	83769	0	54466	54187
	0,2	8	0	42	33	0	0,15	9396	114439	0	83745	0	54467	54187
	0,2	9	0	42	33	0	0,16	9385	114561	0	83769	0	54467	54187
	0,2	10	0	42	33	0	0,17	9375	114561	0	83769	0	54467	54187
Tangier	0,08	8	1	0	0	0	0,07	1124	100000	116176	0	0	29016	65325
	0,08	9	1	0	0	0	0,08	1099	100000	116176	0	0	29016	65325
	0,08	10	1	0	0	0	0,08	1073	100000	116176	0	0	29016	65325
	0,13	8	1	0	0	0	0,08	2575	100000	116176	0	0	29016	65325
	0,13	9	1	0	0	0	0,09	2550	100000	116176	0	0	29016	65325
	0,13	10	1	0	0	0	0,09	2523	100000	116176	0	0	29016	65325
	0,2	8	1	0	0	0	0,09	4606	100000	116176	0	0	29016	65325
	0,2	9	1	0	0	0	0,10	4581	100000	116176	0	0	29016	65325
	0,2	10	1	0	0	0	0,11	4555	100000	116176	0	0	29016	65325

BERKAN	0,08	8	0	47	36	0	0,1	2919	122056	0	84029	0	54329	54179
	0,08	9	0	47	36	0	0,11	2890	122479	0	84709	0	54306	54656
	0,08	10	0	47	36	0	0,12	2879	122479	0	84709	0	54306	54656
	0,13	8	0	47	34	0	0,12	5634	121832	0	84510	0	54313	53894
	0,13	9	0	47	36	0	0,13	5626	122056	0	84029	0	54329	54179
	0,13	10	0	47	36	0	0,14	5595	122479	0	84709	0	54306	54656
	0,2	8	0	48	33	0	0,15	9432	121934	0	84955	0	54297	53835
	0,2	9	0	48	33	0	0,16	9422	121934	0	84955	0	54297	53835
	0,2	10	0	47	37	0	0,17	9438	122005	0	83466	0	54349	53902
DAKHLA	0,08	8	1	0	0	0	0,08	1361	100000	111015	0	0	30475	61632
	0,08	9	1	0	0	0	0,08	1336	100000	111015	0	0	30475	61632
	0,08	10	1	0	0	0	0,09	1309	100000	111015	0	0	30475	61632
	0,13	8	1	0	0	0	0,09	2885	100000	111015	0	0	30475	61632
	0,13	9	1	0	0	0	0,09	2860	100000	111015	0	0	30475	61632
	0,13	10	1	0	0	0	0,1	2833	100000	111015	0	0	30475	61632
	0,2	8	1	0	0	0	0,1	5018	100000	111015	0	0	30475	61632
	0,2	9	1	0	0	0	0,1	4993	100000	111015	0	0	30475	61632
	0,2	10	1	0	0	0	0,1	4966	100000	111015	0	0	30475	61632
Zagora	0,08	8	1	12	8	0	0,12	2950	117975	59817	20955	0	39259	39259
	0,08	9	1	12	8	0	0,13	2928	117757	59817	20739	0	39259	39259
	0,08	10	1	12	8	0	0,14	2898	117757	59817	20739	0	39259	39259
	0,13	8	1	12	9	0	0,14	4936	117975	59817	20955	0	39259	39259
	0,13	9	1	12	8	0	0,15	4914	117776	59817	20827	0	39259	39259
	0,13	10	1	12	8	0	0,15	4886	117757	59817	20739	0	39259	39259
	0,2	8	1	12	8	0	0,16	7718	117854	59817	21035	0	39259	39259
	0,2	9	1	12	8	0	0,17	7697	117770	59817	20790	0	39259	39259
	0,2	10	1	12	8	0	0,17	7663	117814	59817	20966	0	39259	39259
Essaouira	0,08	8	1	14	10	0	0.2	2634	121218	55451	25544	0	40733	40315
	0,08	9	1	14	10	0	0.12	2608	121166	55451	25505	0	40737	40250
	0,08	10	1	14	10	0	0.13	2574	121252	55451	25606	0	40726	40336
	0,13	8	1	14	10	0	0.13	4676	121147	55451	25323	0	40758	40284
	0,13	9	1	14	10	0	0.14	4646	121147	55451	25384	0	40751	40273
	0,13	10	1	14	10	0	0.15	4611	121252	55451	25606	0	40726	40336
	0,2	8	1	14	10	0	0.16	7522	121219	55451	25544	0	40733	40315
	0,2	9	1	14	10	0	0.16	7500	121146	55451	25323	0	40758	40284
	0,2	10	1	14	10	0	0.17	7470	121148	55451	25323	0	40758	40284

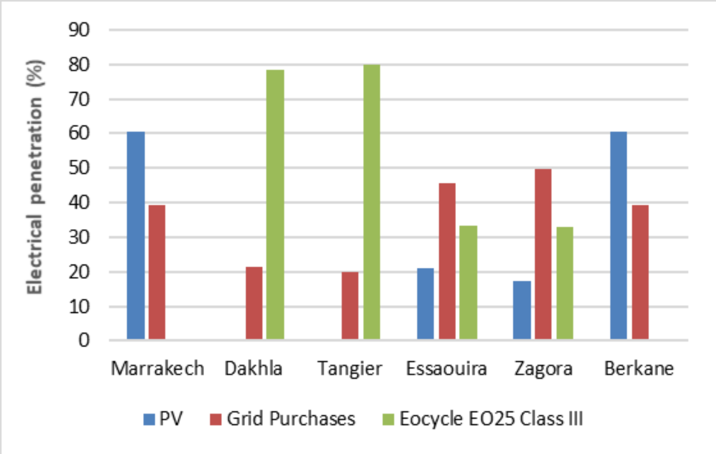


Fig. 9. Annual electricity production by the optimal systems

4.2. Renewable fraction (RF)

The power purchase as well the sell prices have a concrete impact towards the RF. This latter is defined as the fraction of the energy delivered to the load from renewable power sources. HOMER calculates the RF using the following equation:

$$RF = 1 - \frac{E_{nonren}}{E_{served}}$$

Where E_{nonren} is the nonrenewable electrical production and E_{served} is the total electrical energy feeded to meet the load demand. It sould be noted that the sold energy to the grid is included in E_{served}. Fig. 10 shows the RF of the optimal systems in the six cities, the RF has its highest value of 80% in Tangier and Dakhla where the purchased energy (nonrenewable energy) has the lowest value compared to the other cities. The lowest value of the RF is observed in Berkane and Marrakech where the purchased energy is high in these cities. Generally, the analysis of the various systems demonstrated that higher sell-back prices are appreciated in any HRES in grid-connected mode. This could encourage private entities to invest in renewable energies and reach high values of the RF.

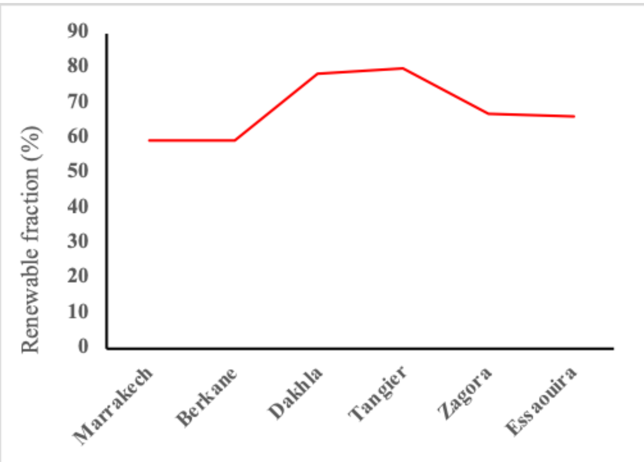


Fig. 10. Renewable fraction of the optimal systems in the different studied cities

4.3. GHG Emission

Different policies towards carbon emissions have been adopted in the intent to deal with climate change. In this context, the European Union Emissions Trading Scheme (EU ETS) has been launched in 1st January 2005 to combat with the greenhouse gas (GHG) emissions cost-effectively. This program is adopted in 31 countries by covering more than 11,000 industrial plants and power plants. A tax can be added which imposes to the producers of electricity to increase the price of electricity in wholesale power markets [51]. This carbon dioxide (CO^2) emissions tax is varying from one country to another, for example this tax is limited in \$1/tons in Mexico, while it is up to \$168/tons CO^2 in Sweden [52]. However, Morocco doesn't adopt any carbon emissions tax until this moment. Fig. 11 illustrates the total emissions generated by the various proposed systems. it can be clearly noted that the highest amounts is registered in Marrakech and Berkane by approximately 35000 kg/year, while Tangier presents the lowest of 18000 kg/year. As illustrated in Fig. 11, the highest carbon emissions can be explained by the dependence on the purchased power that has its maximum value in Marrakech and Berkane.

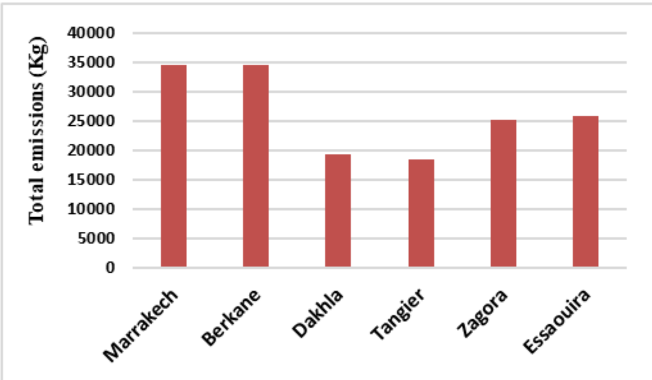


Fig. 11. Total emissions (Kg) produced in the six cities

4.4. Sold and purchased energy

Fig. ***12*** shows the purchased and sold electricity for each city. The maximum sold and minimum purchased energy are registered in Tangier and Dakhla due to their high potential of wind energy as mentioned above. With regard to the other cities, the sold and the purchased energy are almost equal, e.g. in Marrakech and Berkane, the purchased electricity is 55000 kWh while the sold energy is 54000 kWh. The optimal hybrid system in Zagora and Essaouira provides an amount of sold energy of 41000 kWh, while the purchased electricity is limited in 40000 kWh.

4.1. Electricity production for the different studied cities

From Fig. ***13***, the produced energy from the renewable sources based especially on the wind represents more than 87% in Tangier and Dakhla, while the rest is ensured by the grid. As mentioned above, this could be explained by the high potential of those cities in terms of wind energy. For Berkane and Marrakech, the electricity production from the PV represents more than 63 % of the monthly average electric production during the year. An exception was observed in the south eastern region (Zagora city), where the produced energy from PV and wind cover almost 70% and 30% will be provided by the grid.

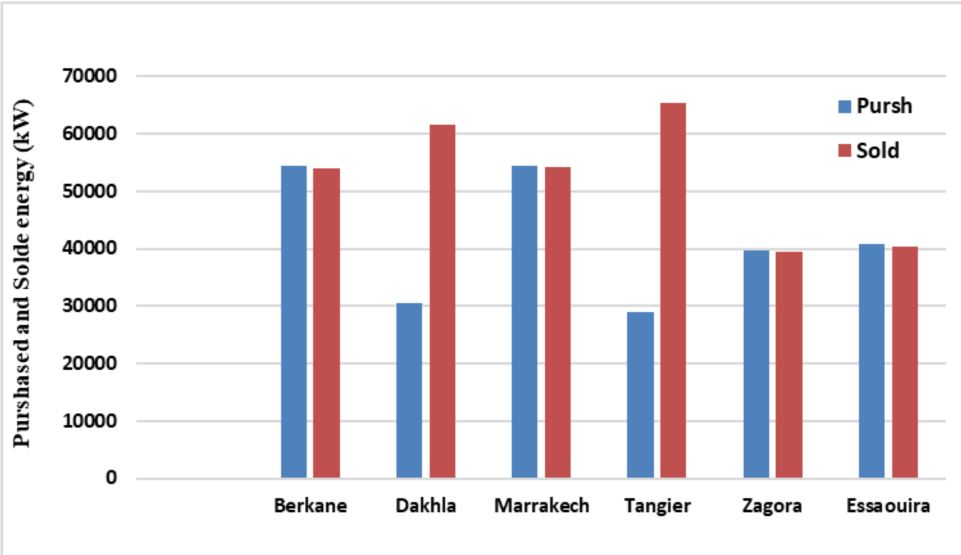
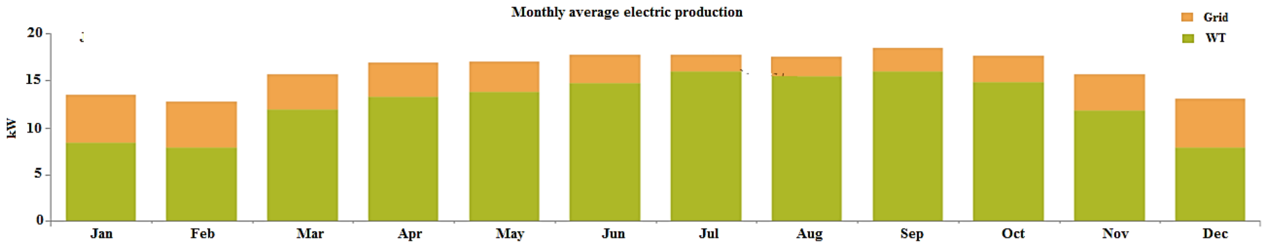
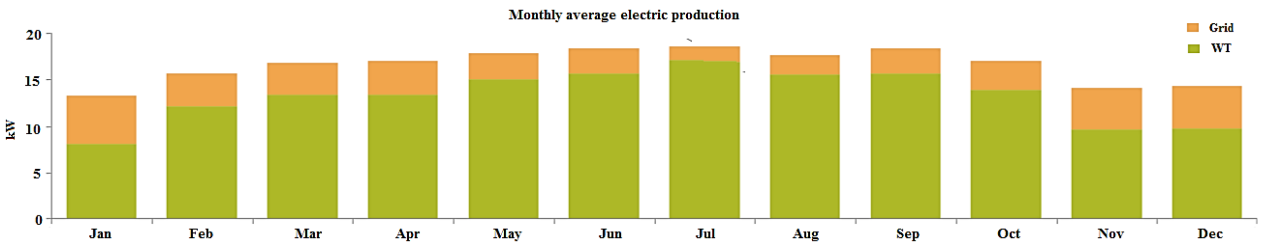


Fig. 12. Purchased and sold energy under the different studied cities

DAKHLA



TANGIER



BERKAN

Fig. 13. The monthly average electric production of PV , wind turbine and grid

4.2. Excess of electricity

The electricity surplus of the optimal systems in each city is presented in Fig. ***14***. It can be observed that all the values of the electricity excess remain limited due to the fact that the systems are grid-connected which allows the integration of the energy excess generated by the renewable sources into the grid. The maximum value of 0.85% is obtained in Berkane, while it is nil in Dakhla and Tangier due to the big amount of the sold energy in those cities.

4.3. Sensitivity analysis

4.3.1. Impact of purchased power price

The grid-connected renewable hybrid system is simulated by considering different purchased power price with a load profile of 219.05 kWh/day and an interest rate of 8 %. In this case study, three different power prices of the purchased energy are considered.

Fig. ***15*** shows the impact of the purchased power price on the COE. It is found that the high purchased power price is translated to a high COE. In fact, the purchased power price ranging from \$0.08 to \$0.2 increases the COE as follows:

- From 0.133 to 0.157 \$/kWh in Berkane,
- From 0.073 to 0.096 \$/kWh in Tangier,
- From 0.111 to 0.157 \$/kWh in Zagora,
- From 0.116 to 0.157 \$/kWh in Essaouira,
- From 0.1 to 0.15 \$/kWh in Marrakech,
- From 0.07 to 0.1\$/kWh in Dakhla.

The impact of the purchased power price has its lowest value in Tangier with 33.3% as increase unlike the other cities with 45%. The difference can be justified by the high amount of the purchased electricity on those cities contrarily to Tangier.

4.3.2. Impact of the interest rate

Fig. ***16*** illustrates the effect of the interest rate on the COE for a purchased power price of \$0.08/kWh. An increase of the IR from 8% to 10% increases the COE. As shown in this figure, Tangier city has the lowest value of the COE compared to the other cities. The highest value is observed for the cities of Zagora where the COE is ranging from $0.14 to $0.15 per kWh. Meanwhile, it is found that for an IR increase of 2%, the COE increases by 16.06% in Marrakech and Berkane and by 7% in the other cities. It should be noted that even with an IR of 10%, the COE did not exceed $0.1 per kWh in Tangier and Dakhla. This is almost equal to the price of conventional electricity from the national electrical grid 0.1 $/kWh (social tariffs).

4.3.3. Impact of the wind speed and power price

The site characteristics where the power plant is installed could be influenced by many factors such as trees, buildings and mountains. The local wind speed may be relatively different from that recorded by the weather station. On the other hand, the purchased power price has an effect on the HRES design. Therefore, a sensitivity analysis is considered in this case study in order to assess the impact of wind speed and purchased power price variation. Fig. 17 and Fig. 18 show the data map (DMAP) diagram. This latter is used to determine the optimal system in the six cities with a wind speed ranging from 4 to 8 m/s and a power price ranging from 0.08 to 0.2$/kW. From Fig. ***17***, it is observed that the system having wind turbine combination produces optimal results in wind speed values ranging from 4 and 8 m/s, whereas the based solar combination is able to produces optimal result in lower wind speed. It is observed from the same figure that the PV/WT/Grid is considered as the optimal hybrid system for all the wind speed values ranging from 4 to 8 m/s in Tangier, while in Dakhla, Essaouira and Marrakech, the PV/Grid become optimal for a wind speed low than 5.40 m/s and a power price of 0.1 $/kWh. In Zagora the PV/WT/Grid occupied the big margin and it is considered as the optimal system for all wind speed values exceeding 4.7 m/s and a power price of 0.1 $/kWh.

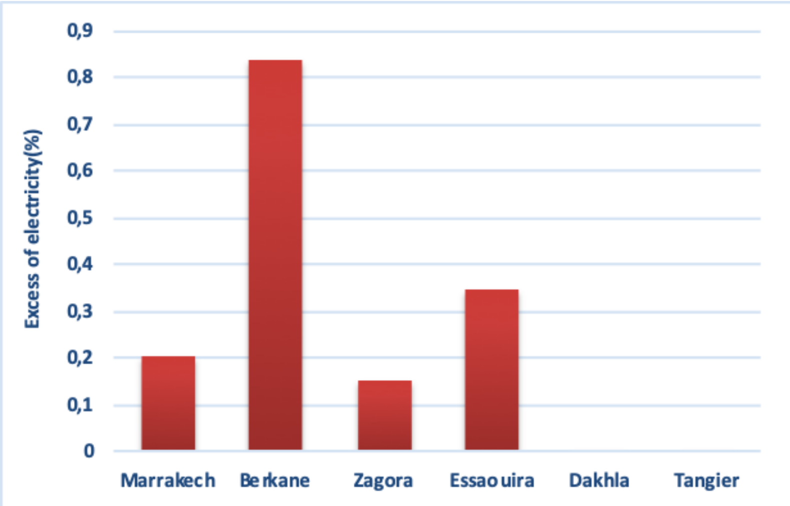


Fig. 14. Electricity surplus registered by the six cities

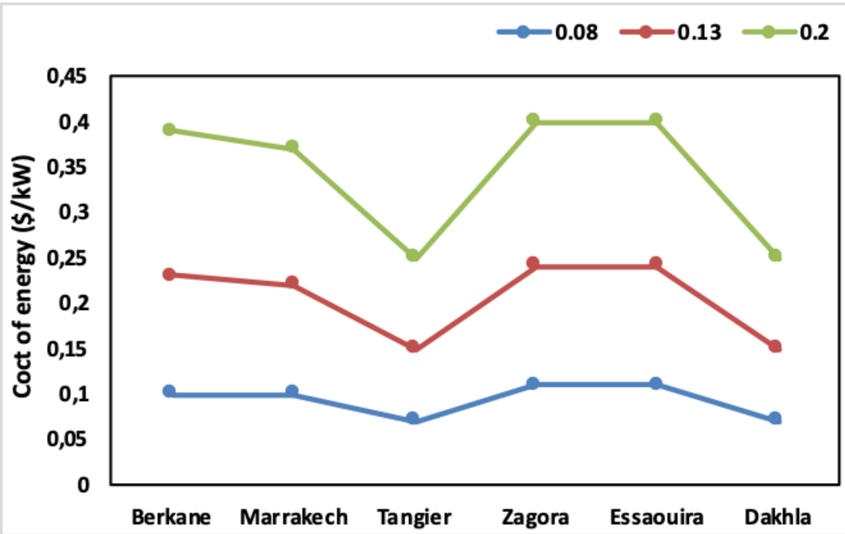


Fig. 15. Impact of the power price on the COE

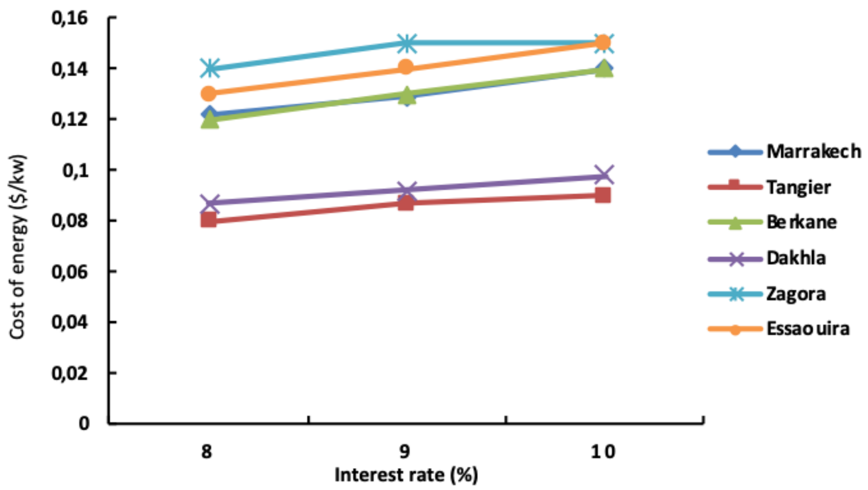


Fig. 16. Impact of the interest rate on the COE

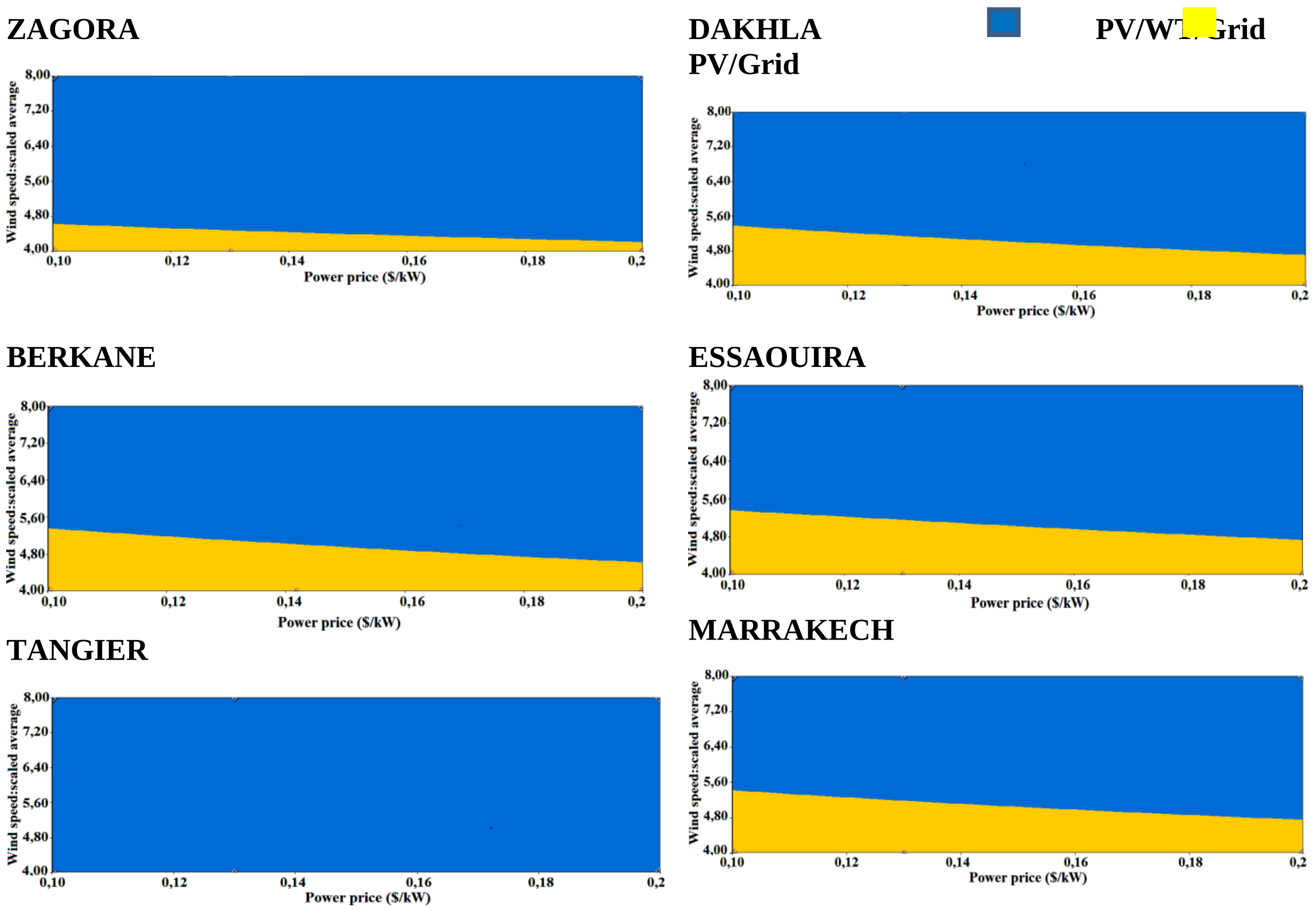


Fig. 17. Effect of wind speed and power price on optimal configuration

From an economic point of view, the system cost parameters of the optimal configurations for all developed cases are reviewed in Fig. *18*. The annual electric production through surface plotted based on the NPC as represented by the colored band. From this surface plot, it is found that the NPC is influenced by the wind speed and the power price. In fact, the NPC decreases when the wind speed increases and it is proportional to the power price increase. The high NPC is registered in Tangier with $145 305.6 for a power price of 0.2 $/kW and wind speed of 4m/s.

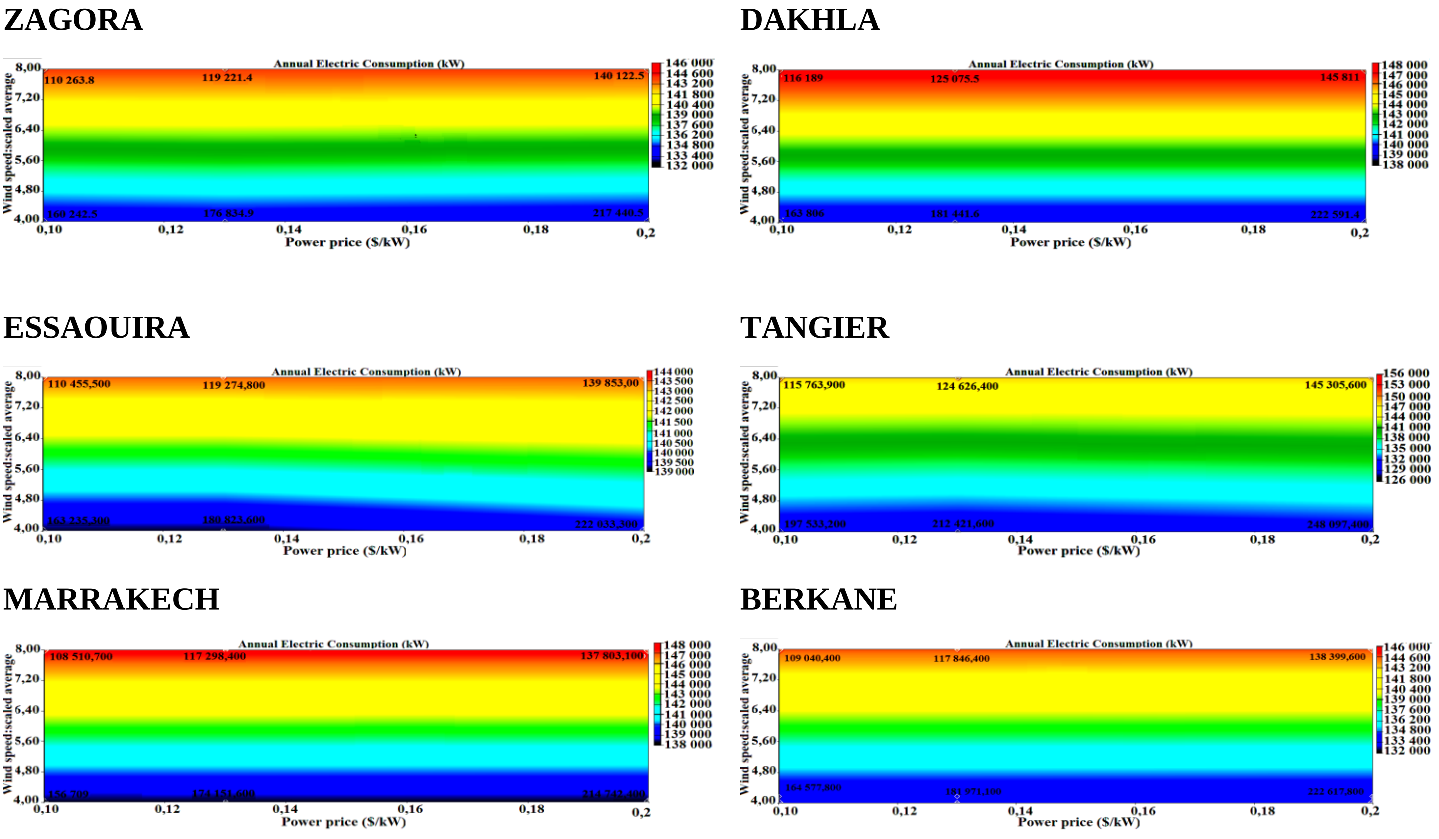
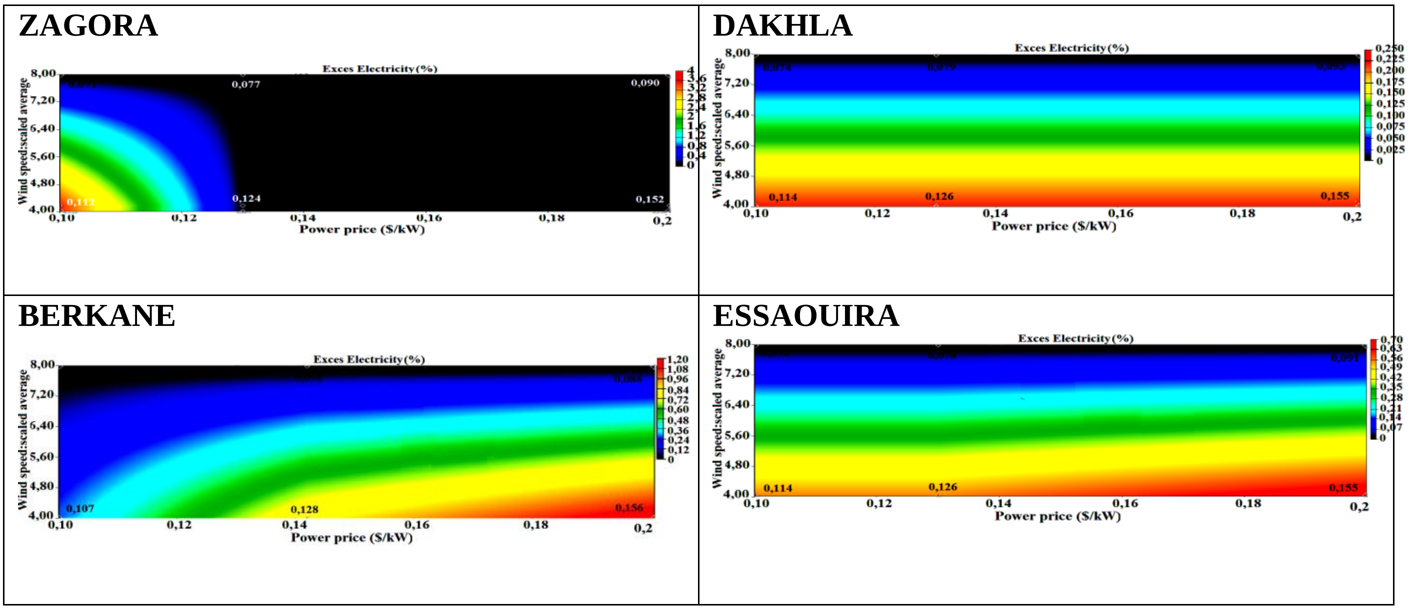


Fig. 18. Dmap of the wind speed and energy price on the annual electric production

Fig. ***19*** shows the excess of electricity represented by the colored band superimposed by the COE. From this surface plot, it is found that the excess of electricity has its maximum in Tangier and Berkane, while its minimum is observedFig. in Dakhla. For a wind speed of 4 m/s and different energy prices, the optimal system has a high COE and a high energy surplus compared to other cities. However, when the wind speed exceeds 5m/s, the COE becomes the lowest one with 0.073, 0.079 and 0.092$/kW. Moreover, the whole presented systems in the different studied cities, presented a respectable amount of electricity excess which does not exceed a maximum of 0.1 and 0.2 %. It must be also noted that from a wind speed less than 5 m/s, the annual production has almost the same amount in the six cities. Whereas, with a wind level between 5.6 m/s and 8 m/s, a considerable increase is noted in Tangier (156 000 kW) followed by Dakhla (148 000 kW).



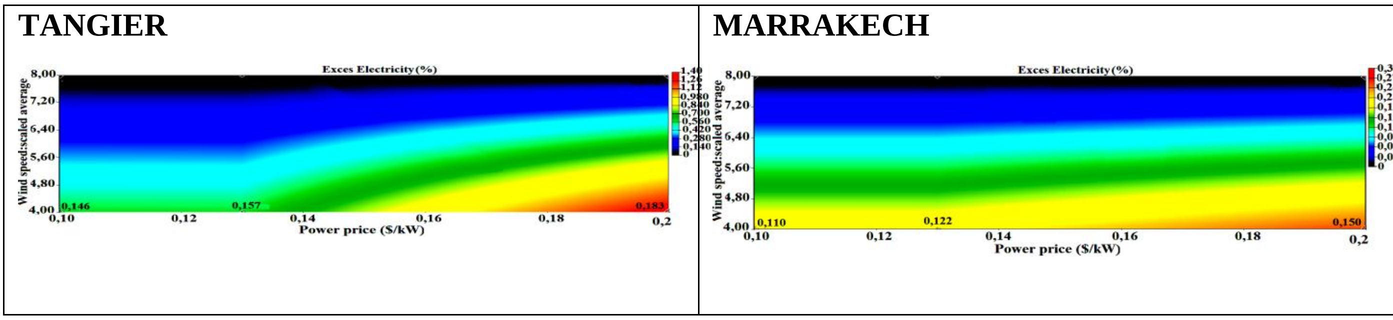


Fig. 19. Dmap of the wind speed and power price on the excess of electricity

5. Conclusion

The work presented a detailed technical and economic feasibility of HRES to supply the energy demand of typical infrastructures including public lighting, governmental institutions, and school campus. The study was carried out in various regions in Morocco to investigate the implementation of small and medium Microgrid project based on wind and solar resources. In fact, HOMER Pro software was used to provide optimal and suitable solutions in the different cities. Several parameters such as interest rate, wind speed and purchased power price were analyzed to evaluate their impact on the viability of the proposed systems. The results show that the based wind systems are more suitable in the cities located in the west of Morocco, while in the eastern cities, it is more suitable to use PV energy. It was also found that a combination of PV and wind energy is more convenient in the cities of ZAGORA and Essaouira. Generally, the obtained results demonstrated the economic viability of the proposed systems in the six cities. This is due to the fact that the obtained COE is comparable with the proposed COE of the national Office of Electricity.

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